Anastasiia Iosimovska

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Summary

Second-year MSc student in Materials Science at Skoltech. Research interests include atomistic modeling, computational materials discovery, and structure–property relationships in materials. Experienced in first-principles calculations, crystal structure prediction, and scientific data analysis.

Education

MSc, Skolkovo Institute of Science and Technology — **Sep 2024 – present**
Major: Materials Science — Moscow, Russia
GPA: 5.00/5.00

BSc, Lomonosov Moscow State University — **Sep 2019 – Jun 2023**
Major: Geology — Moscow, Russia
GPA: 4.74/5.00

Additional education

Workshop, Higher School of Economics — **Nov 2025**
Program: Machine Learning Methods for Materials Property Prediction and Generation — Moscow, Russia

Course, Lomonosov Moscow State University — **Nov 2022**
Program: Artificial Intelligence in Chemistry: Rational Design and Automated Discovery — Moscow, Russia

Professional Retraining Diploma, Skolkovo Institute of Science and Technology — **Oct – Nov 2022**
Moscow, Russia
Program: Digital Modeling of Advanced Thermoelectric Materials

Research experience

MSc Researcher, Industry-Oriented Computational Discovery Laboratory, Skolkovo Institute of Science and Technology — **Sep 2024 – present**
Moscow, Russia
Supervisor: Prof. Alexander G. Kvashnin
MSc research project
Project: Computational Study of Ammonia Adsorption and Decomposition on W–B Nanoclusters
Methods: DFT calculations, adsorption-energy analysis, NEB reaction-path calculations, automated processing of VASP outputs, and structural visualization.

Intern, Artificial Intelligence Research Institute — **Jun – Aug 2025**
Moscow, Russia
Project: Machine-Learning Potentials for Accelerated Modeling of W–B Nanoparticles

Student Researcher, Materials Discovery Laboratory, Skolkovo Institute of Science and Technology — **Sep 2023 – Sep 2024**
Moscow, Russia
Supervisor: Prof. Artem R. Oganov
Project: Thermodynamic Stability and Ionic Conductivity in Li/Na/K–Ge Binary Systems
Methods: Crystal structure prediction, first-principles calculations, thermodynamic stability analysis, ionic conductivity analysis, and scientific visualization.

Student Researcher, Laboratory of New Materials for Solar Energy, Lomonosov Moscow State University — **Jun – Aug 2023**
Moscow, Russia
Supervisor: Dr. Ekaterina I. Marchenko
Project: Theoretical Assessment of Chloride Ion Influence on Grain Growth of Hybrid Perovskites
Methods: interatomic-potential modeling, grain-boundary stability calculations, atomistic simulations, structural analysis, and data analysis.

Research skills

Computational methods: DFT, crystal structure prediction, molecular dynamics, phonon calculations.
Software: VASP, Quantum ESPRESSO, FHI-aims, ORCA, Gaussian, MOPAC, LAMMPS, USPEX, Phonopy.
Programming and data analysis: Python, Bash, automated processing of computational outputs, scientific visualization.
HPC and documentation: Linux-based HPC workflows, LaTeX, MS Office.
Visualization: VESTA, OVITO, ChemCraft, Jmol.

Publications

1) Iosimovska, A. V., Maltsev, A. P., Chepkasov, I. V., & Oganov, A. R. (2024). Thermodynamic stability and ionic conductivity in lithium–germanium binary system. *Applied Physics Letters*, *124*(16).
2) Maltsev, A. P., Iosimovska, A. V., Chepkasov, I. V., & Oganov, A. R. (2025). Structure transformations and ionic conductivity in germanides of sodium and potassium. *Journal of Materials Chemistry A*, *13*(20), 14612-14619.
3) Marchenko, E. I., Belich, N. A., Iosimovska, A. V., Misyutin, V. A., Goodilin, E. A., & Tarasov, A. B. (2024). Theoretical assessment of chloride ion influence on grain growth of hybrid perovskites. *Mendeleev Communications*, *34*(3), 321-324.
4) Chepkasov, I. V., Baidyshev, V. S., Iosimovska, A. V., Zamulin, I. S., & Kvashnin, A. G. (2025). Adsorption properties of crystalline and amorphous PdIr nanoparticles. A systematic first-principles study. *Journal of Catalysis*, *447*, 116102.

Conference participation

Oral presentations

1. 68th All-Russian Scientific Conference at MIPT, Section “Modern Methods of Atomistic Modeling in Materials Science”. Moscow, Russia. 2026.
2. 67th All-Russian Scientific Conference at MIPT, Section “Modern Methods of Atomistic Modeling in Materials Science”. Moscow, Russia. 2025.
3. Lomonosov Student Conference. Lomonosov Moscow State University, Moscow, Russia. 2023.

Poster presentations

1. Youth Conference of the Institute of Biochemical Physics, Russian Academy of Sciences and Universities “Biochemical Physics”. Moscow, Russia. 2024, 2025.
2. 2nd Sino-Russian Symposium on Chemistry and Materials. 2024.
3. Modern Trends in Computational Materials Discovery, 21st USPEX Workshop. Isfahan University of Technology, Isfahan, Iran. 2022. Poster presenter and tutor.

Awards and Honours

1. Presidential Scholarship for Students Enrolled in Higher Education Programs.
2. Scholarship for Development and Academic Excellence, Skolkovo Institute of Science and Technology.
3. Top Paper Scholarship, Skolkovo Institute of Science and Technology.
4. Winner’s Certificate, Day of Scientific Creativity at Lomonosov Moscow State University.
5. Best Poster Award, 2nd Sino-Russian Symposium on Chemistry and Materials. 2024.

Languages

English (upper-intermediate/academic working proficiency), **Russian** (native)